

Syllabus – Software Engineering – CSC 4700 Sections 001 and 002 – Spring 2026

Course Meeting Times: Section 001: TR 10:00 – 11:15 a.m. Section 002: TR 11:30 a.m. – 12:45 p.m.

Modality: In-person

Location: Mendel G92

Prerequisite: CSC 1052 or ECE 2620

Blackboard: All course materials are in the course sections “Spr26_CSC_4700_001 (or _002)”

Class Session Recordings: All sessions will be recorded and made available via a link on Blackboard.

Instructor: Ms. Kristin Obermyer, Associate Instructor, Dept. of Computing Sciences

Office Hours: Mondays 3:25 – 4:25 p.m. and Thursdays 1:45 – 2:45 p.m. in Mendel 167A.

Instructor and TA office hours are also available by appointment with 24 hours’ advance notice.

Communication: Professional, respectful, and concise communication is expected. Response times will be approximately 24 hours on weekdays, and next business day on weekends/holidays.

- The course Microsoft Teams site is the primary mechanism to communicate with the instructor, TA and classmates.
- Email is secondary and should be used judiciously, such as to provide an absence excuse. Copy the TA on all emails to the instructor. kristin.obermyer@villanova.edu
- Complex issues are best addressed during the instructor’s or TA’s office hours.

Teaching Assistant (TA): Ms. Julia Foy jfoy3@villanova.edu

TA Office hours: To be announced

Course description: Management and production of software systems; the software life cycle; software design techniques and methodologies; participation in a team software development project.

Course objectives

- Survey the discipline of software engineering, including development processes, life-cycle models, quality issues, requirements analysis, design techniques, testing, and project management.
- Study and practice oral and written communication skills with respect to software development projects.
- Explore modern software development tools.
- Gain a strong enough foundation in software engineering that it could lead to a career.

Required text: A combination of relevant articles and sections of **free** eBooks will be used. The eBooks will be accessed via the O’Reilly for Higher Education portal, which is provided for free by Falvey Library: <https://go.oreilly.com/villanova-univ>. The eBooks can be accessed via browser or the O’Reilly Media app for Android and iOS. Students are expected to come to class prepared to discuss the assigned reading, and a reading guide will be provided for each topic. Core free eBooks and a key website are listed below:

- Winters, Titus, et al. *Software Engineering at Google*. O’Reilly, 2020.
- Brooks, Frederick P. *The Mythical Man-Month*. Anniversary Ed., 2nd Ed. Addison-Wesley, 1995.
- Cohn, Mike. *Learn About Agile*. Mountain Goat Software website.
<https://www.mountaingoatsoftware.com/agile/new-to-agile-or-scrum>.

Hardware/Software Requirements: Students may use any computing device and operating system that’s available to them. A laptop and high-speed Internet connection are recommended to complete assignments. Bring your device to class. All software used in this course is free. Key software will include Blackboard, Microsoft Teams, Poll Everywhere, Respondus Lockdown Browser, Java SE, Eclipse or VS Code IDE, Git/GitHub, and Lucidchart.

Class Session Structure: A typical class session will start with an overview of the agenda and an interactive review from the prior class. Interactive lectures will be interspersed with hands-on activities. Discussion of the assigned reading will be done in both small groups and with the full class.

Course requirements for assessment of students' learning: Final grades will be determined based on the following areas and associated weighting factors. Each area is described in detail below the table. Any questions about grades on any item must be addressed within one week of the item being graded.

Area	Weighting Factor
Team Project (subject to Individual Contribution Factor)	40%
Two Exams	35%
Homework	20%
Engagement and Professionalism	5%

Assessment Areas: Different types of assessments are used in this course to evaluate student progress toward course objectives. All assessments are intended to deepen students' understanding of the course topics and to provide hands-on experience and expert feedback. Some assessments will be graded in detail, but others may be graded in a less granular fashion or will be graded solely for completion. It's possible for both individual and team assessments to be due on the same day, since the class only meets twice per week.

- **Team Project:** Small teams will use an agile approach to complete a semester-long software project of their own choosing, within scope and complexity parameters specified by the instructor. The project will include the following key aspects: envision a project concept, elicit and analyze requirements, design and diagram an initial solution, develop and evaluate a prototype. Project assignments will be due approximately weekly. Students will be given opportunities to provide feedback on their teammates' contributions, and lack of contribution will result in an individual's project score being downgraded. Final presentations of the project will be held during the last two weeks of classes and will replace the final exam.
 - **Individual Contribution Factor (ICF):** For the semester project, the team overall grade, which is the sum of multiple deliverables, is multiplied by the **Individual Contribution Factor (ICF)** when determining an individual's project grade. The ICF is determined based on peer evaluations and instructor observation. Earning 100% for the ICF means that the student contributed equally to the project, and therefore earns the full team score. **Failure to contribute equally will result in a lower ICF, which will reduce the student's share of the team score, and thus their final grade.**
- **Two Exams:** Two 75-minute exams will be given during the semester. Exams will be taken in-person using the Respondus Lockdown Browser for Blackboard, and will be closed notes and closed book.
- **Individual Homework:** Assignments will include technical writing, analysis of case studies, programming and unit testing.
- **Engagement and Professionalism:** Engagement with the in-class exercises is imperative. In-class work may include design exercises, discussions, debates, games, and brainstorming. Consistent attendance, on-time arrival, professionalism in interactions and communications with the instructor, TA, classmates and guest speakers, and avoidance of digital distractions during class sessions all contribute to this aspect of the grade.

Course Outline and Calendar: Available on Blackboard as "Course Schedule" and will be updated regularly. The Course Schedule indicates dates of exams, presentations and assignment due dates, plus learning goals for each class session. The course is organized into a series of learning modules focused on specific aspects of software engineering. Key dates for this semester:

Exam #1: Thurs. February 19

Exam #2: Tues. March 31

Project Presentations: Tues. April 28

Grading scale: The following grading scale will be used to determine final letter grades in the course, based on the weighted average for all areas of the course. Grades are not “rounded up” and must cross the minimum threshold without rounding to earn the associated letter grade.

Grade Range	Grade Letter		Grade Range	Grade Letter
93.0 – 100.0	A		73.0 – 75.9	C
90.0 – 92.9	A-		70.0 – 72.9	C-
86.0 – 89.9	B+		66.0 – 69.9	D+
83.0 – 85.9	B		63.0 – 65.9	D
80.0 – 82.9	B-		60.0 – 62.9	D-
76.0 – 79.9	C+		<= 59.9	F

Attendance and Lateness Policies:

- Attendance at all classes is mandatory, and attendance will be taken at the beginning of each class.
- If you have a valid reason to miss a class, such as illness or using a Personal Day, contact the instructor via email as soon as possible to explain your absence. Only absences covered by the [University’s list of excused absences](#) will be excused.
- Any absences beyond two excused absences will be considered unexcused unless written documentation is provided within one week of the absence.
- Arrive on time for class, and maximize your work time with your team. Arriving late and/or repeatedly exiting/re-entering during the session are disruptive to the instructor and your fellow classmates. If you arrive 10 or more minutes late, you will be marked late. Excessive unexplained absence from the classroom during the class session will be marked late as well. Three lates will be considered equivalent to one unexcused absence.

Quality Standards for Written Work: All written work must be proofread and subjected to spelling and grammar checks prior to submission. Any written work that doesn’t meet these standards will be penalized 10% of the assignment grade and returned for correction. Further, all sources must be properly cited using MLA citation format, and direct quotations must be cited with double quotes and an inline reference. **Failure to properly cite sources will result in an academic integrity violation.**

Use of Artificial Intelligence (AI): The use of generative AI (gen AI) in this course is constrained **by type of artifact**, as described below. Any usage outside specified parameters will result in an academic integrity violation. Students are responsible for any errors produced by AI assistance, and AI will not be available during exams. When gen AI is used, the raw prompts and the AI’s responses must be submitted with each release demo. Use the AI vendor’s built-in tools to maintain a record of your interactions, and then submit a link to it with your demo artifacts. Examples of generative AI include, but are not limited to, ChatGPT, Gemini, Claude, DeepSeek, and Copilot. University AI Guidelines for students are available at [this link](#).

AI guidelines by type of artifact:

- **Project** - Projects may integrate with generative AI using vendor APIs to create AI-infused applications.
- **Code/Tests/Data** - Gen AI may be used for assistance with code, configuration, debugging, unit tests, and sample data. However, **you are responsible for making sure the code executes correctly**. You must be able to explain every line of code in your project, regardless of how it was created. The instructor and TA will not debug large chunks of AI-generated code.

- **Written content** – The use of gen AI is **not permitted** for written content in this course. Written content includes user stories, slides and speaker’s notes, reports, self/peer evaluations, and emails. Learning to communicate technical information in a precise and concise manner is part of this course.
- **Architectural Diagrams** – The AI features of Lucidchart may be used to assist with architectural diagram creation, but you are responsible for using the correct diagrammatic notation. In terms of architecture and design choices, it’s better to consult with your instructor about options that will best meet your project goals.
- **Wireframes** – The AI features of Figma may be used to assist with the creation of wireframes for your project.
- **Images** – AI image generators such as Stable Diffusion may be used to create icons and images for your application and to illustrate your project presentation. Be sure to give credit appropriately for generated images.
- **Feedback** – Your instructor will provide expert feedback on all artifacts. We will experiment with AI feedback as well.
- **Citations** – You may use AI to generate properly-formatted citations for content (articles, videos, podcasts) that you have actually read or viewed. You may not use AI to fabricate or enhance a list of citations.
- **Learning new topics** – Gen AI may be used to help you learn new technologies and concepts. However, it’s best to start with the official tutorials and documentation provided by the vendor.

Late assignment submissions: Late assignments are subject to 10% per day lateness penalty, and cannot be accepted more than 10 days late. Requests to submit late work without penalty (such as due to illness) must be made prior to the due date via email to the instructor and TA, and evidence of work on the assignment must be presented in order to qualify for late submission. Each student is automatically granted a “one day late pass” which can be used to have one extra day to submit one assignment for any reason. To use a late pass, simply note it in your assignment submission on Blackboard.

Regrades: Regrades of resubmitted assignments are not available unless expressly indicated by instructor in the initial grade for the work. Improvements are always welcomed for the student’s educational benefit.

Extra credit opportunities: No extra credit opportunities are currently scheduled for the course.

Collaboration: It is strongly recommended that each student attempt to do the individual assignments on their own first, before coming to office hours or consulting with others. Working through challenging assignments is a key part of learning. You may discuss the nature of the assignment and general approaches with other students in the course, provided submitted work clearly states with whom and the degree to which students collaborated. When collaborating with others in this manner, make sure it is an equitable situation that enhances, rather than replaces, your own learning. You should NOT share code, diagrams or written work with others for individual assignments, and you should not submit a copy or variation of another student’s work as your own.

Missed Exams: Missed exams may not be made up without prior approval of the instructor. Students must provide a valid, documented reason that an exam must be missed, prior to the scheduled start time of the exam. Exams may not be made up or taken early solely for the convenience of students, such as for travel plans. If you must miss an exam for a valid reason (such as a medical emergency), it is your responsibility to contact the instructor as soon as possible, ideally before the exam.

Peer Tutoring: Peer tutoring by Computing Sciences students is available via The Learner's Studio. See this link for more information: <http://www.csc.villanova.edu/help/>.

University Policies: *The policies in the following sections are provided by Villanova and apply to all courses.*

Personal Day: In addition to the attendance policy stated above, students are entitled to two excused absences for any reason that may contribute to their personal wellness. Students must advise the instructor by email before class of their intent to utilize a Personal Day as the reason for their absence. A Personal Day will not be approved retroactively. Students may, but are not required, to provide additional information regarding their absence. Additionally, Personal Days may not:

- be used on consecutive class days;
- be used in the same week;
- be used immediately preceding or following a University holiday or break period;
- be used on days when exams, presentations or other major assignments are scheduled.

A Personal Day does not grant an automatic extension for items due. Students remain responsible for all assignments, exams, presentations, etc. due on that date. It is in the instructor's discretion to determine whether any extension is appropriate given individual circumstances. For more information see [Villanova University's Class Attendance Policy](#).

Office for Access and Disability Services (ADS) and Learning Support Services (LSS): It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. All students who need accommodations should go to [Clockwork for Students](#) via myNOVA to complete the Online Intake or to send accommodation letters to professors. Go to the [LSS website](#) or the [ADS website](#) for registration guidelines and instructions. If you have any questions please contact LSS at 610-519-5176 or learning.support.services@villanova.edu, or ADS at 610-519-3209 or ods@villanova.edu.

Academic Integrity: All students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. You may view the [University's Academic Integrity Policy and Code](#) for a detailed description. Additional helpful information is available via Falvey Library's [Academic Integrity Gateway](#). The [Dept. of Computing Sciences Academic Integrity Policy](#) extends the University Policy with computing-specific information.

If a student is found responsible for an academic integrity violation, which results in a grade penalty, they may not WX the course unless they are approved to WX for significant medical reasons. Students applying for a WX based on significant medical reasons must submit documentation and their request for an exception will be considered.

Absences for Religious Holidays: Villanova University makes every reasonable effort to allow members of the community to observe their religious holidays, consistent with the University's obligations, responsibilities, and policies. Students who expect to miss a class or assignment due to the observance of a religious holiday should discuss the matter with their professors as soon as possible, normally at least two weeks in advance. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the absence. More information can be found at this link:

<https://www1.villanova.edu/villanova/provost/resources/student/policies/religiousholidays.html>

Copyright and Intellectual Property: The materials used in Villanova University courses ("Course Materials") generally represent the intellectual property of course instructors, third parties and/or the University which may not be disseminated or reproduced in any form for public distribution (e.g., sale, exchange, etc.) without the written permission of the course instructor. Course Materials include all written or electronic documents and materials, including syllabi, current and past examination questions/answers, and presentations such as lectures, videos, PowerPoints, etc., provided by a course instructor. Course Materials may only be used by students enrolled in the course for academic (course-related) purposes. Published course readings (book chapters, articles, reports, etc.) available in Blackboard are copyrighted material. These works are made available to students through licensed databases or fair use. They are protected by copyright law, and may not be further disseminated or reproduced in any form for distribution (e.g., uploading to websites, sale, exchange, etc.) without permission of the copyright owner. More information about intellectual property, copyright, and computer acceptable use can be found here: <https://www1.villanova.edu/villanova/generalcounsel/copyright.html>

Inclement Weather: In the event of inclement weather, we will follow the Villanova policy documented at this link: <https://www1.villanova.edu/villanova/provost/resources/policies/snow-policy.html>.

Class Session Recordings: Class sessions will be recorded by the instructor for educational purposes and available to the class during the semester. Students may access the recorded class sessions via Blackboard. The recordings will allow students to review content after class and provide an opportunity for students who are unable to attend class to view the class session. Students are not permitted to make their own recordings of class sessions, including office hours. Students must not distribute or post the recordings for any reason. Students requesting the use of assistive technology as an accommodation should contact Learning Support Services and/or the Office for Access and Disability Services.